3.1.1 Laws of reflection

The incident ray, reflected ray and normal lie in one plane. The angle of reflection equals the angle of incidence, with both measured from the normal. On a curved or rough surface, use the local normal at the incidence point.

On a smooth surface, similarly directed normals keep parallel incident rays parallel after reflection, allowing a clear image. On a rough surface, the normals vary, so rays reflect in different directions. This is diffuse reflection, and each ray still obeys the laws of reflection.

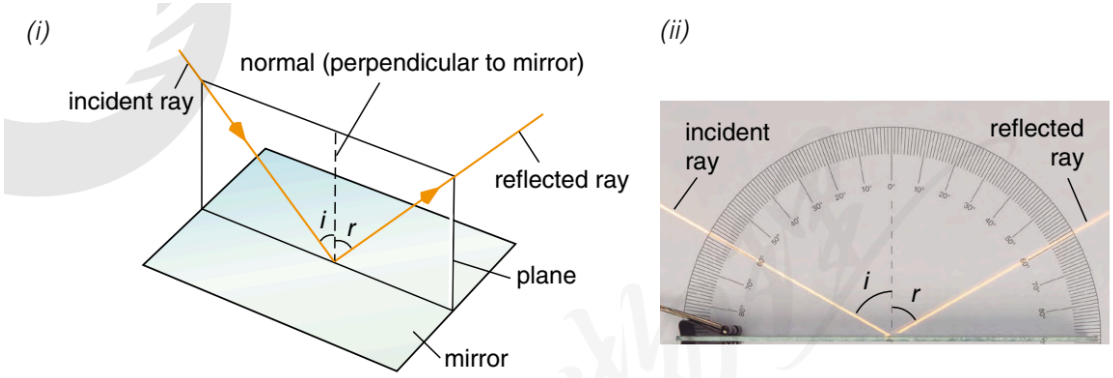


Fig 1.2c *The laws of reflection.*

The incident ray, reflected ray and normal share one plane. Both i and r are measured from the normal, giving i = r.

3.1.2 Images in a plane mirror

A plane mirror forms an upright, same-sized, laterally inverted virtual image. Its perpendicular distance behind the mirror equals the object's perpendicular distance in front. Reflected rays do not actually pass through the image point. Only their backward extensions meet there, so the image cannot be projected onto a screen.



Fig 1.2j *The image formed by a mirror is laterally inverted.*

The original photograph shows lateral inversion. The virtual image appears behind the mirror, with the same size and perpendicular distance as the object.